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Properties of Materials
Lecture 11

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Exercise Seminar 1
Lecture 8: Bonding Between Atoms, Learning Objectives, Bonding Energies, Primary Bonding, Ionic Bonding, Covalent Bonding, Metallic Bonding, Crystal Structures, Body-Centred Cubic, Face-Centred Cubic, Secondary Bonding, Van der Waals Forces, Hydrogen Bonding, Bonding Energies and Properties, Bonding Types and Properties, Concepts and Terminology (4)
Lecture 9: Recap from Last Time, Task
Exercise Seminar 2 / Lecture 10: Packing of Atoms in Solids (1), Learning Objectives, The Structure of Metals and Ceramics (1), Unit Cell Parameters, Face-Centred Cubic (fcc), Hexagonal Close-Packed (hcp), Crystal Structures - Metals, Crystal Structures - Ceramics, Concepts and Terminology (5)
Lecture 11: Packing of Atoms in Solids (2), Learning Objectives, Atomic Packing, Atomic Packing - bcc Structures, Atomic Packing Factor - bcc, Atomic Packing - fcc Structures, Atomic Packing Factor - fcc, Atomic Packing - hcp Structures, Atomic Packing Factor - hcp, Tips for Calculating APF for hcp, Concepts and Terminology (6)

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Packing of atoms in solids

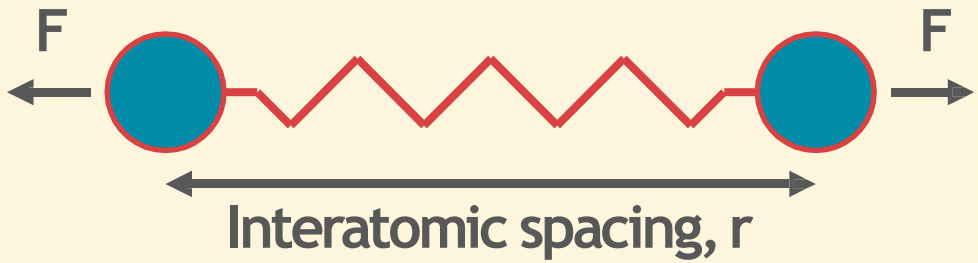
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Recap from last time...

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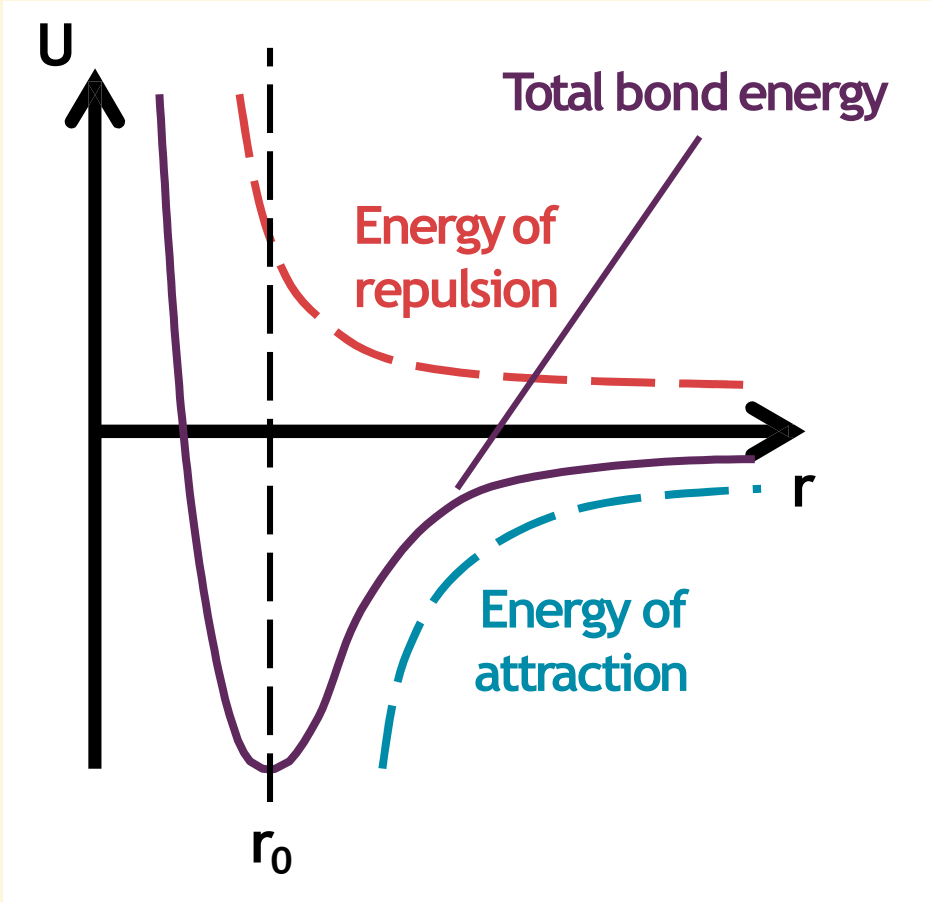
Bonding energies (1)

- Ball and spring model:
  - Atoms subjected to competing repulsive and attractive forces



- At rest, atoms will be in their lowest energy state

- r_0 = equilibrium interatomic spacing
  U = potential energy of bond



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The structure of metals and ceramics (2)

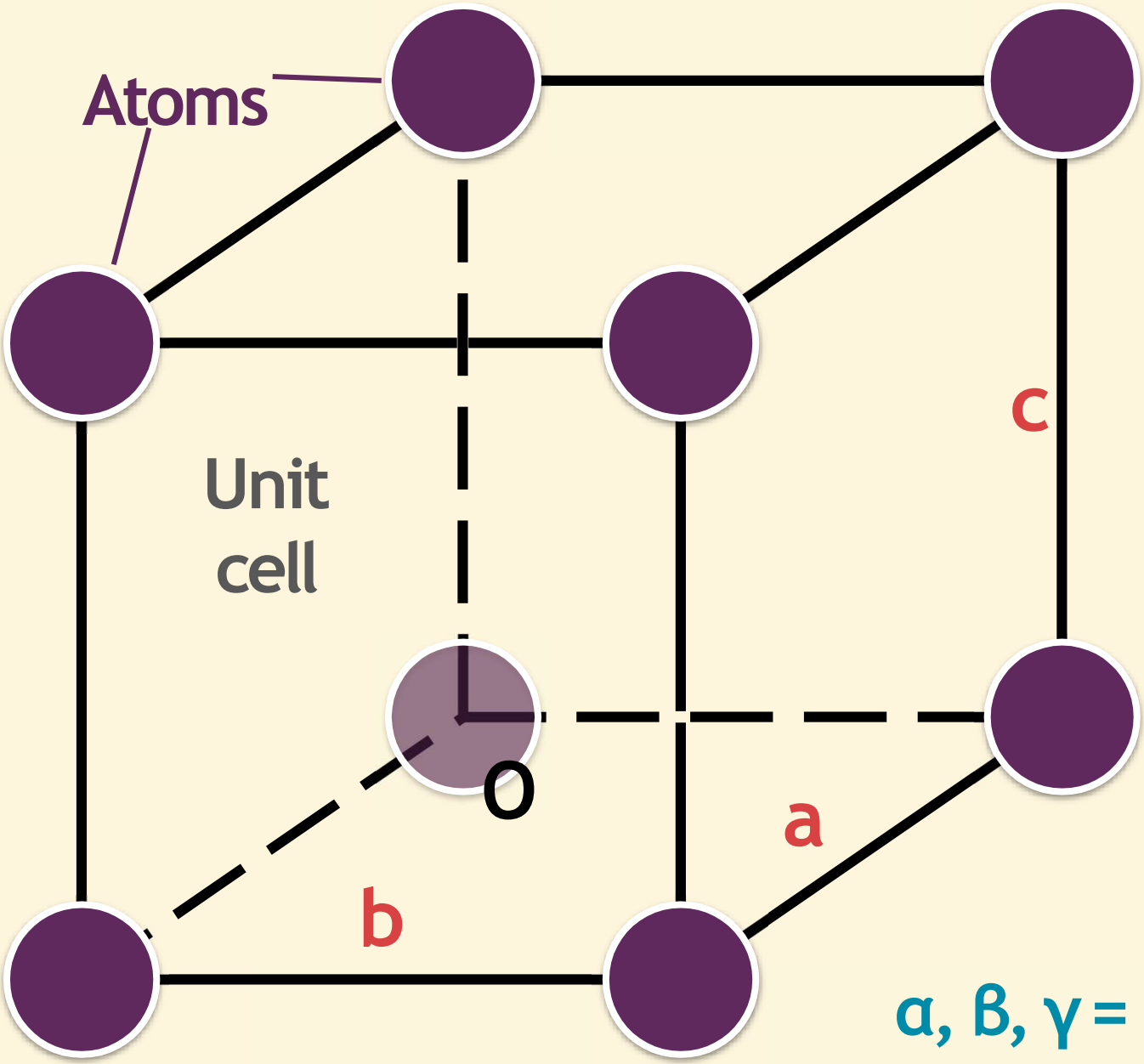
S6

S5

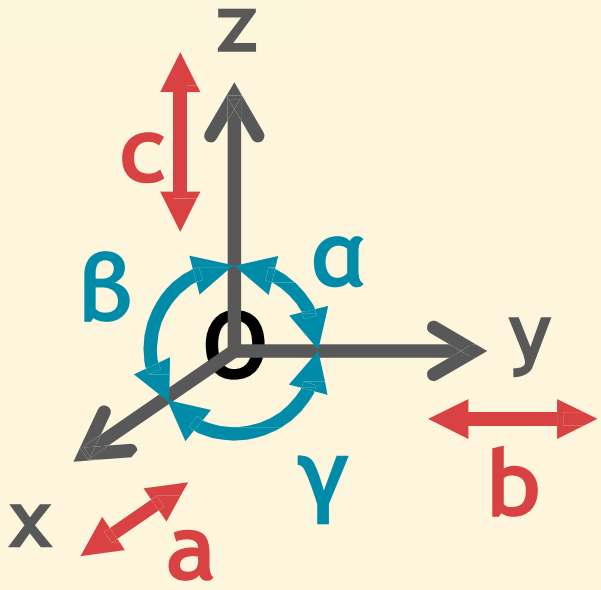
S4

- Metals/ceramics are made up of a number (~10^{15} m^{-3}) of **grains**
- *Grains are typically imperfect and are themselves typically made up of a number of **sub-grains***
- Each grain can be considered as a **crystal**; an ordered configuration of atoms
- Each crystal contains a number (>10^{9}) of **unit cells** with the same lattice structure and orientation
- Each unit cell contains a number of main / primary / **substitutional atoms** that are connected by **primary bonds**
- *Each unit cell may also contain a number of smaller secondary/ **interstitial atoms** in between larger atoms*

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Simple cubic structure

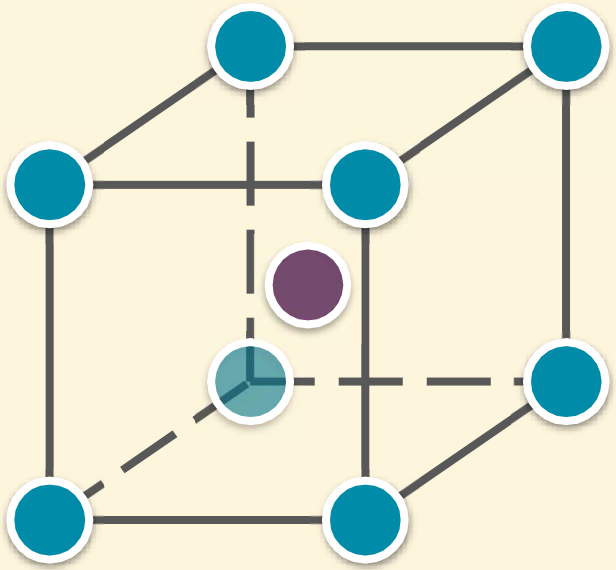


a, b, c = lattice parameters

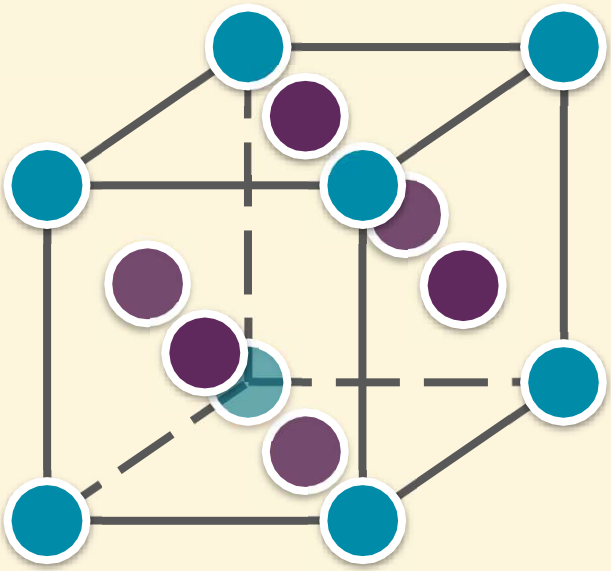
α, β, γ = lattice angles

Unit cell parameters

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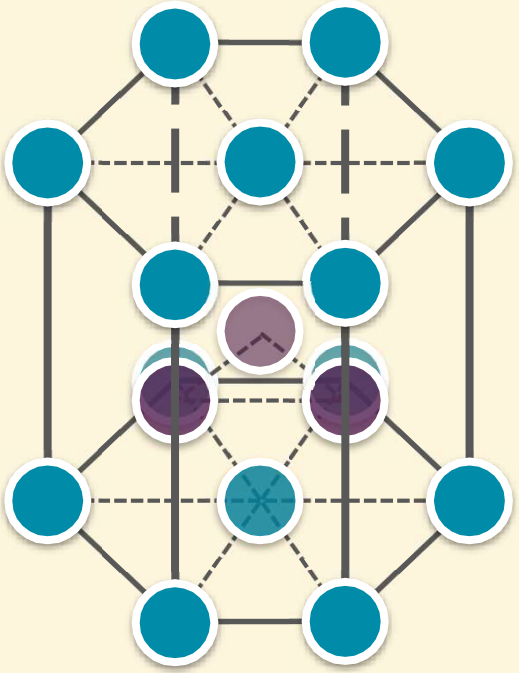


Strong / hard materials



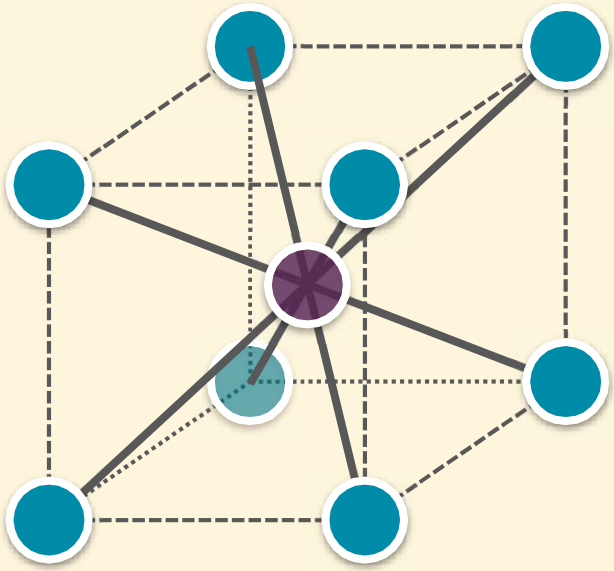
Ductile materials

(Some) light materials



Crystal structures – Metals

- **bcc**: Tungsten (W), Chromium (Cr), α-Iron (ferrite)
- **fcc**: Copper (Cu), Aluminium (Al), Nickel (Ni), γ-Iron (austenite)
- **hcp**: Zinc (Zn), Titanium (Ti), Magnesium (Mg)

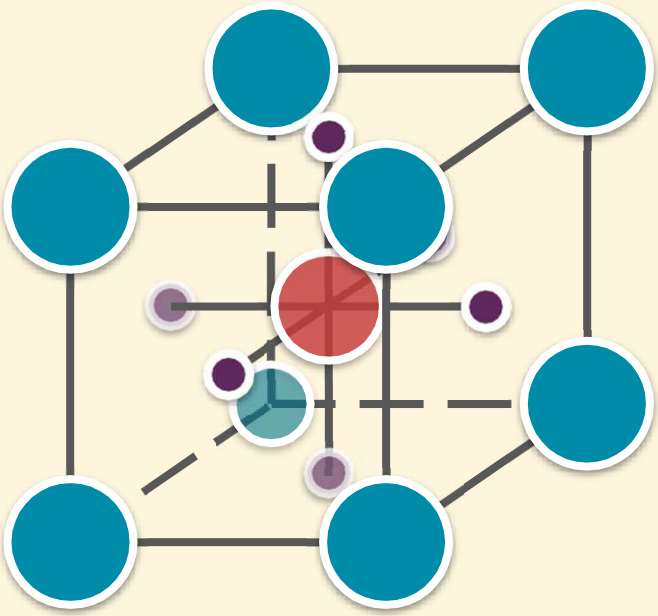


Caesium Chloride

$Cs^+ + Cl^- \longrightarrow CsCl$

Calcium Titanate

$Ca^{2+} + TiO_3^{2-} \longrightarrow CaTiO_3$



Crystal structures – Ceramics

- Ceramics have far more **complex** structures
- With a mixture of **ionic** and **covalent** bonding (high bond energies) they are very difficult to deform

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Learning objectives

- To learn how to characterise the ‘density’ of different atomic arrangements by their **atomic packing factor**

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Atomic packing

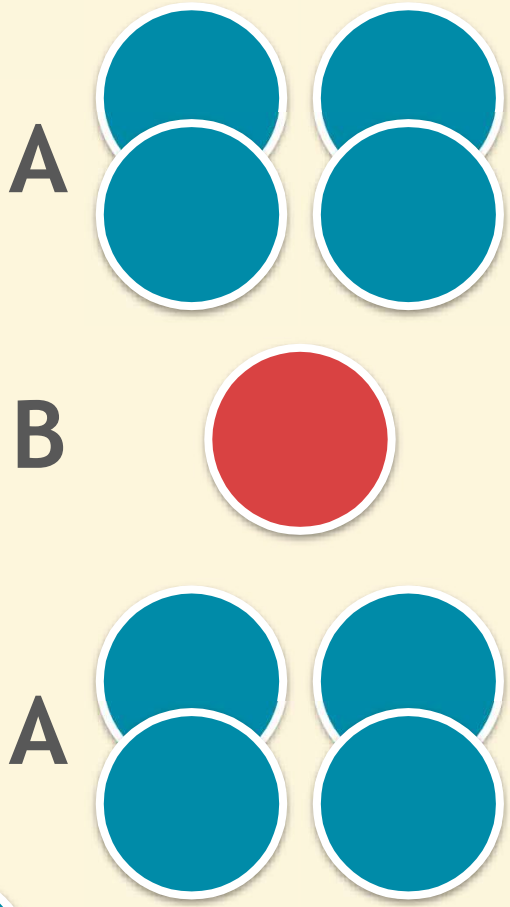
- The arrangement of atoms within the crystal structure has a significant influence on their properties and behaviour
- Specifically, the arrangement of atoms determines how many **slip planes** there are and therefore how easily slip may occur

- Atomic packing therefore describes:
  - The **arrangement of alternate layers of atoms** within the crystal structure
  - A measure of **how 'full' of atoms the unit cell** is, or the atomic density of the material
  - Can be characterised by the **atomic packing factor**,
  - APF = $\frac{N_{atoms}V_{atoms}}{V_{unit\,cell}}$

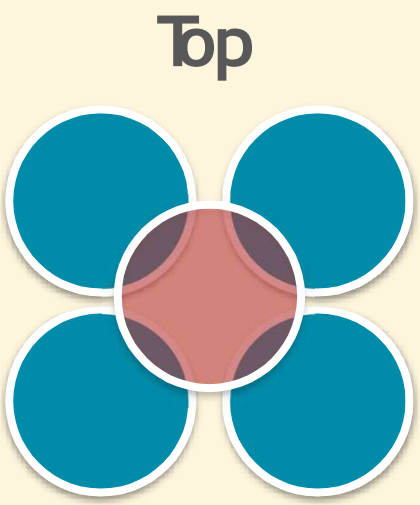
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Atomic packing – bcc structures

- Two layers of four atoms (A) sandwich layers of single atoms (B)



- **Atoms along a line passing through the centre atom 'touch'**
- Atoms in alternate 'A' layers do not touch



- Each atom has 8 neighbours

Atomic packing factor – bcc

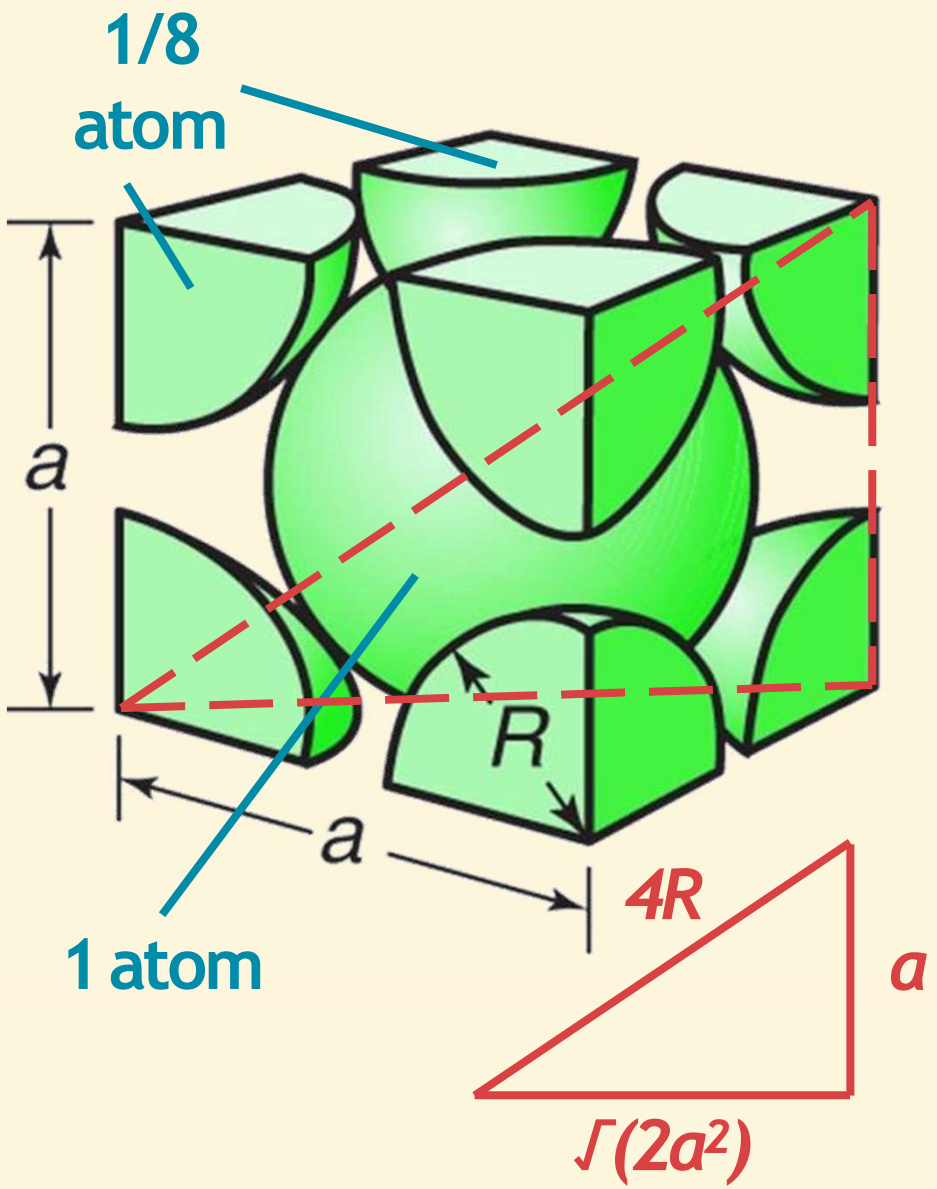
$$\text{APF} = \frac{N_{\text{atoms}} V_{\text{atoms}}}{V_{\text{unit cell}}}$$

$$N_{\text{atoms}} = 8\left(\frac{1}{8}\right) + 1(1) = 2$$

$$V_{\text{atoms}} = \frac{4}{3}\pi R^3$$

$$V_{\text{unit cell}} = a^3 = \left(\frac{64}{3\sqrt{3}}\right) R^3$$

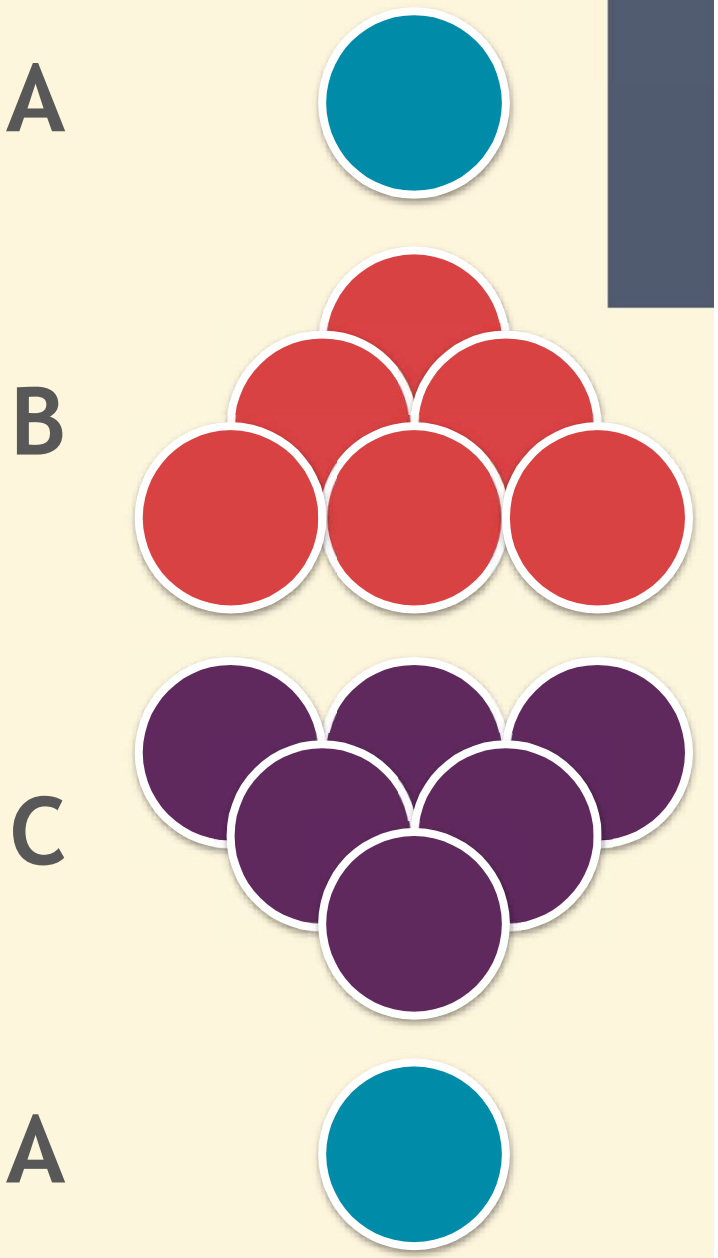
$$\therefore \boxed{\text{APF} = \frac{\pi\sqrt{3}}{8} = 0.68}$$



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Atomic packing – fcc structures

- Single atoms (A) lie in the centre of alternating layers of 6 atoms (B, C)
- **Atoms touch all adjacent atoms**
- Each atom has 12 neighbours



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Atomic packing factor – fcc

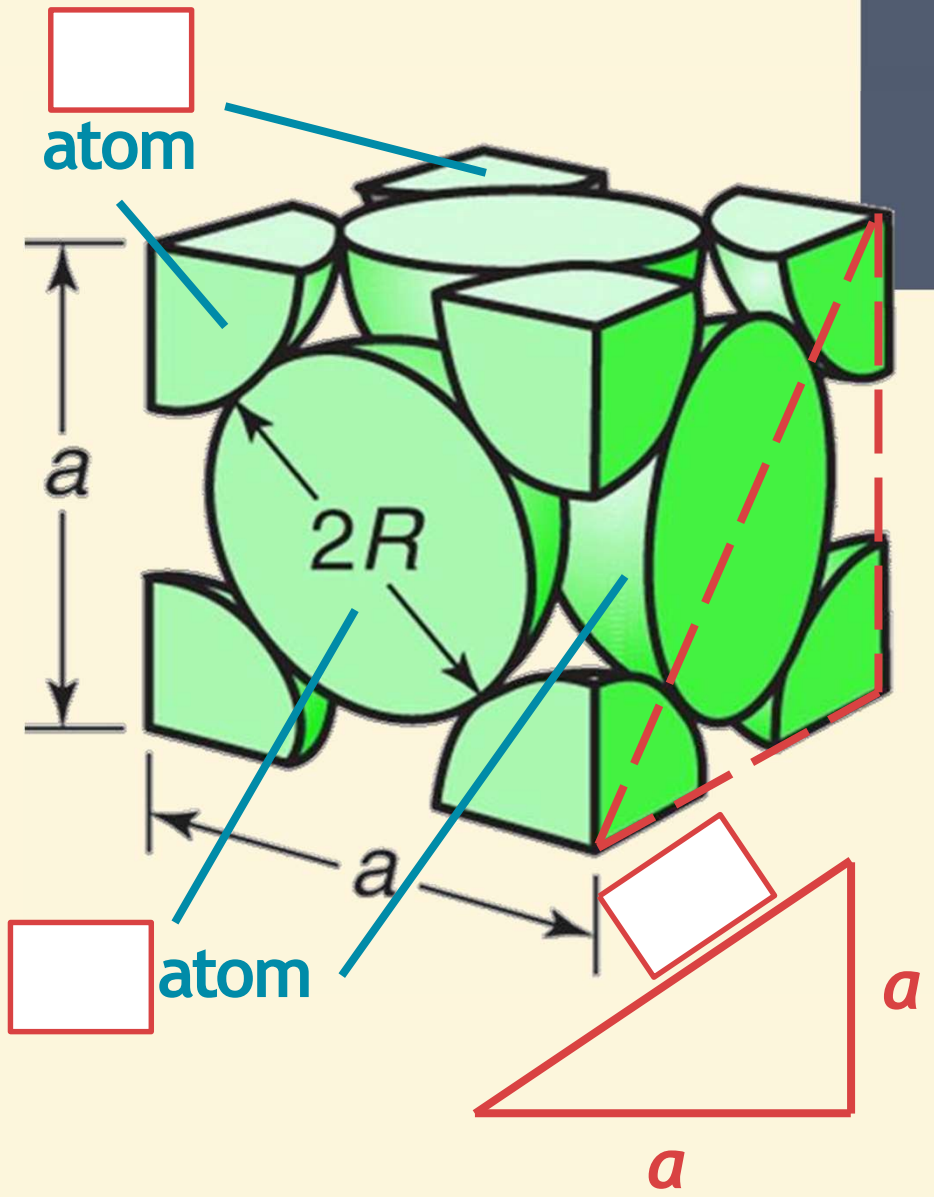
$$\text{APF} = \frac{N_{\text{atoms}} V_{\text{atoms}}}{V_{\text{unit cell}}}$$

$N_{\text{atoms}} =$ [ ]

$V_{\text{atoms}} = \frac{4}{3}\pi R^3$

$V_{\text{unit cell}} = a^3 =$ [ ]

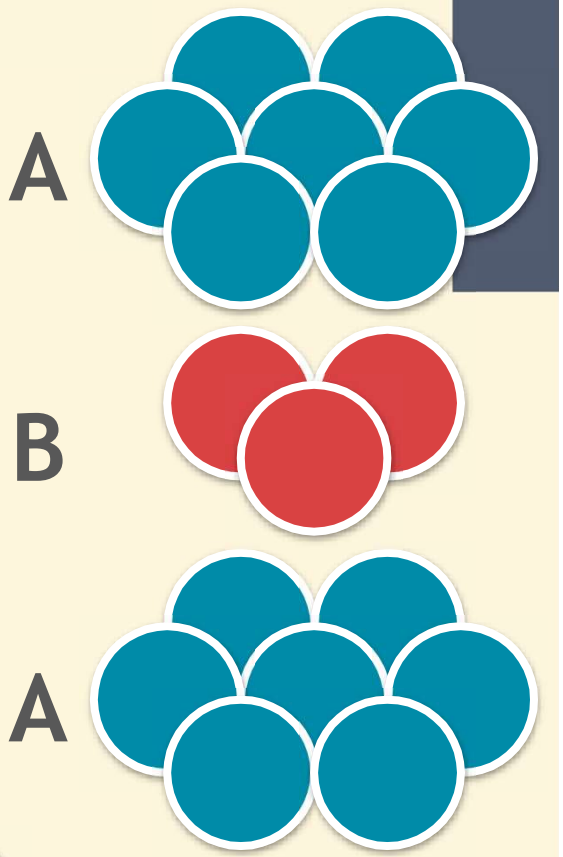
$\therefore$ APF = [ ]

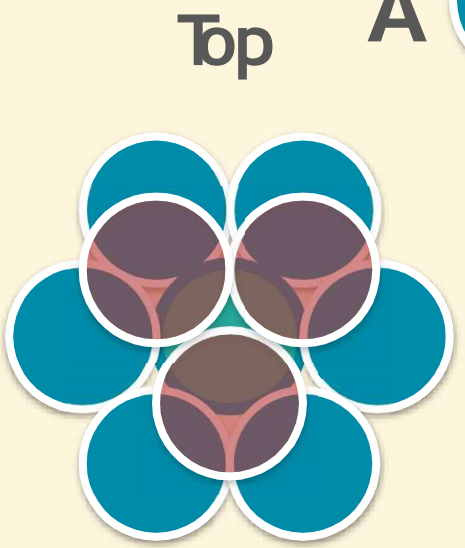


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Atomic packing – hcp structures

- Three atoms (B) lie in the centre of alternating layers of 7 atoms (A)
- **Atoms touch all adjacent atoms**
- Each atom has 12 neighbours





Atomic packing factor – hcp (1)

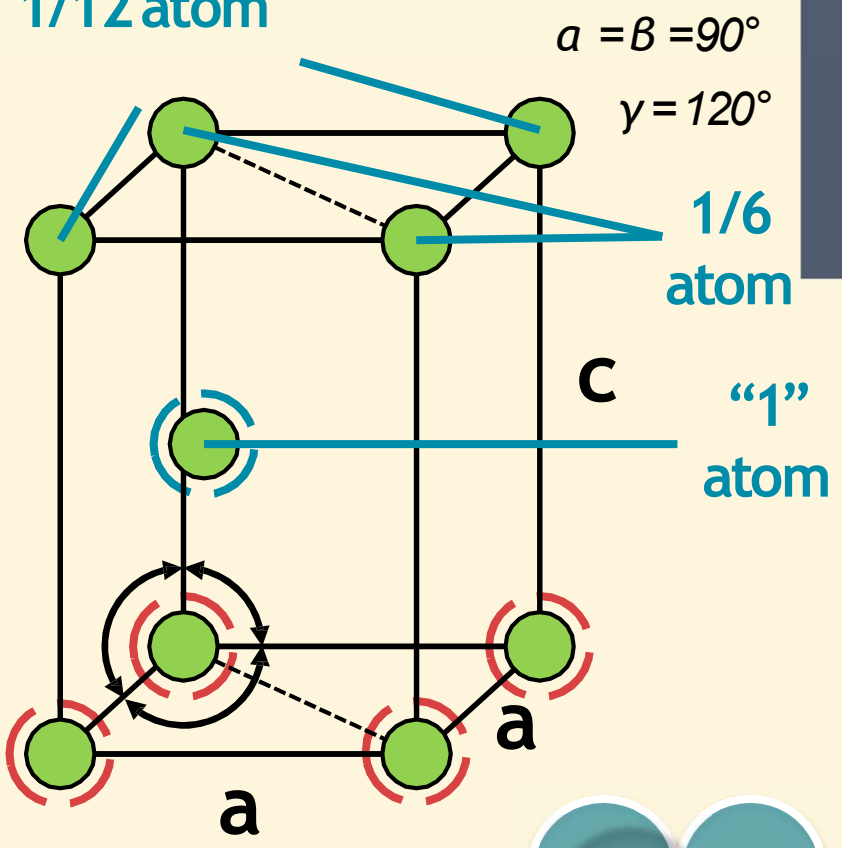
$$APF = \frac{N_{atoms} V_{atoms}}{V_{unit\ cell}}$$

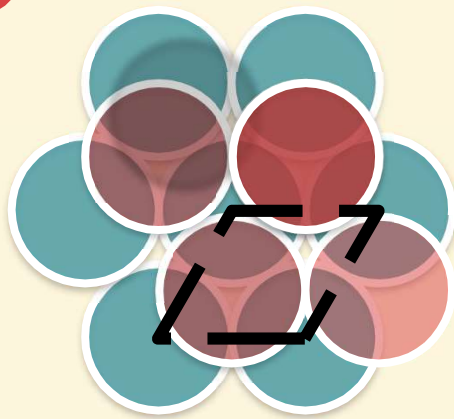
$$N_{atoms} = 4(1/12) + 4(1/6) + 1(1) = 2$$

$$V_{atoms} = \frac{4}{3}\pi R^3$$

$$V_{unit\ cell} = A_{base}(c) = a^2 \cos(30°) c$$

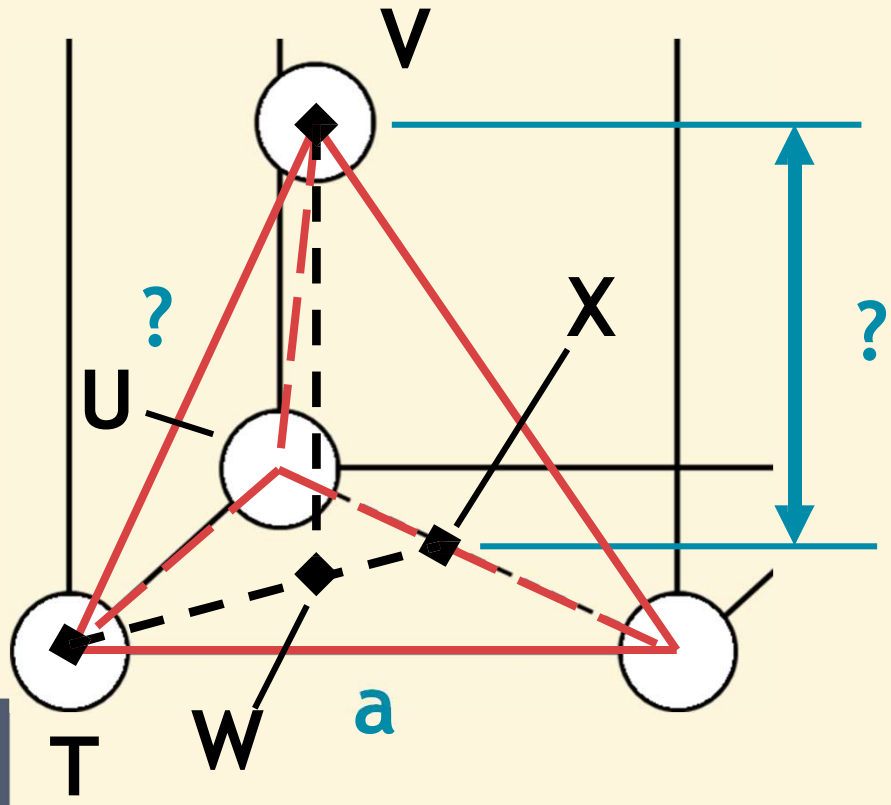
1/12 atom



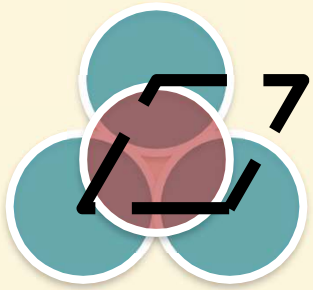


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Atomic packing factor – hcp (2)



$a = \beta = 90°$
$\gamma = 120°$



$$APF = \frac{N_{atoms} V_{atoms}}{V_{unit\ cell}}$$

$$N_{atoms} = 2$$

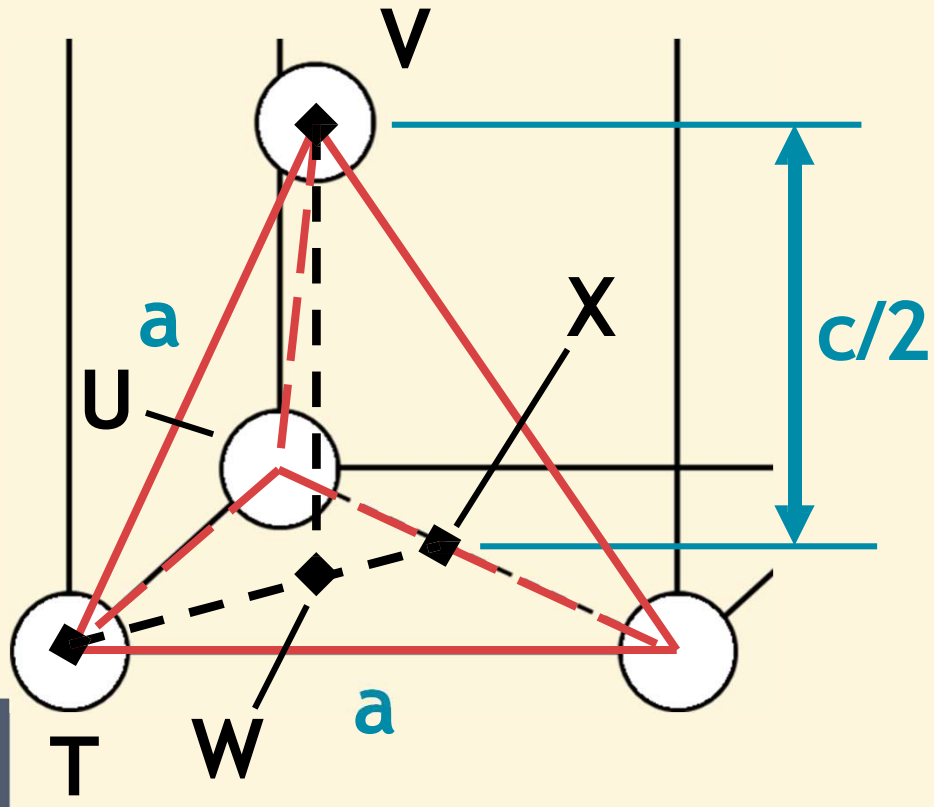
$$V_{atoms} = \frac{4}{3}\pi R^3$$

$$\overline{WV} = ?$$

Distance between **T** and **V**?

$$\overline{TV} = ?$$

Atomic packing factor – hcp (3)



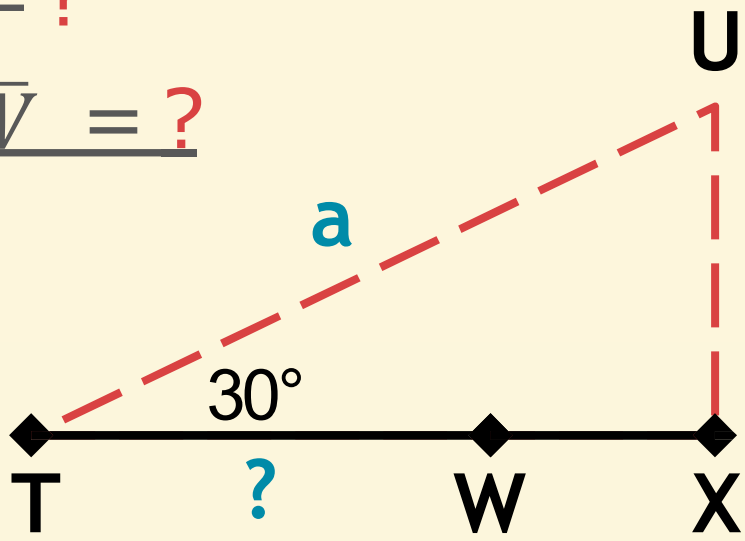
$\overline{WV} = \frac{c}{2}$; $\overline{TV} = a = 2R$

Find **TW**:

$\overline{TW} = ?$

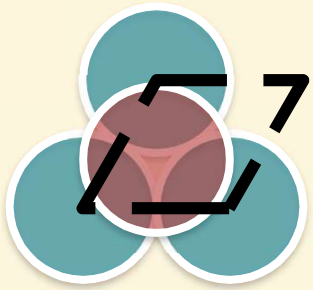
$\overline{TX} = ?$

$\therefore \overline{TW} = ?$

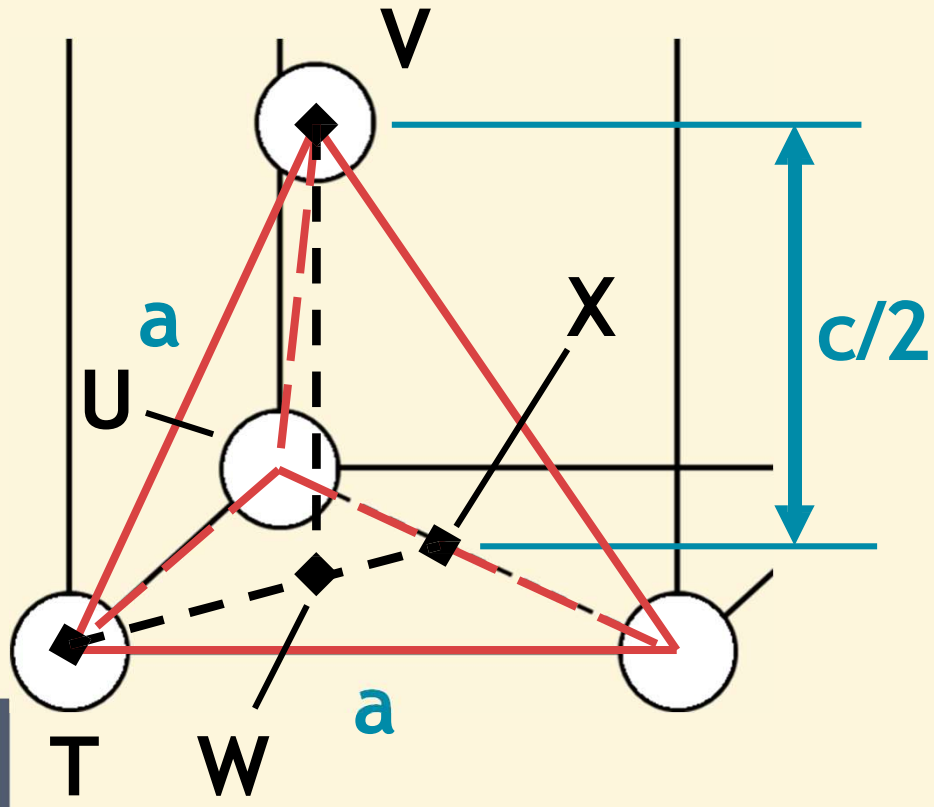


$a = \beta = 90°$

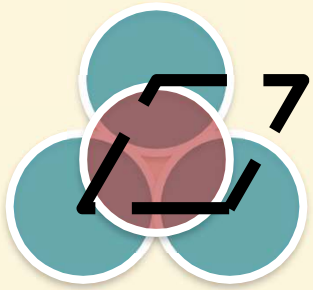
$\gamma = 120°$



Atomic packing factor – hcp (4)



$a = \beta = 90°$
$\gamma = 120°$



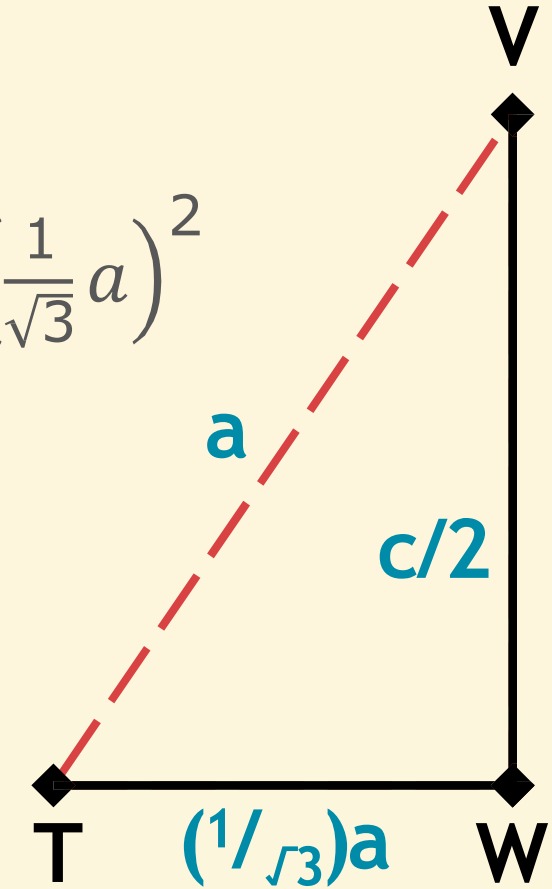
$$\overline{WV} = \frac{c}{2} \; ; \; \overline{TV} = a = 2R$$

$$\overline{TW} = \frac{1}{\sqrt{3}}a$$

$$\left(\frac{c}{2}\right)^2 = a^2 - \left(\frac{1}{\sqrt{3}}a\right)^2$$

$$\Rightarrow c = \sqrt{\frac{8}{3}}a$$

$$\therefore c = 4\sqrt{\frac{2}{3}}R$$



Atomic packing factor – hcp (5)

$$N_{atoms} = 2$$

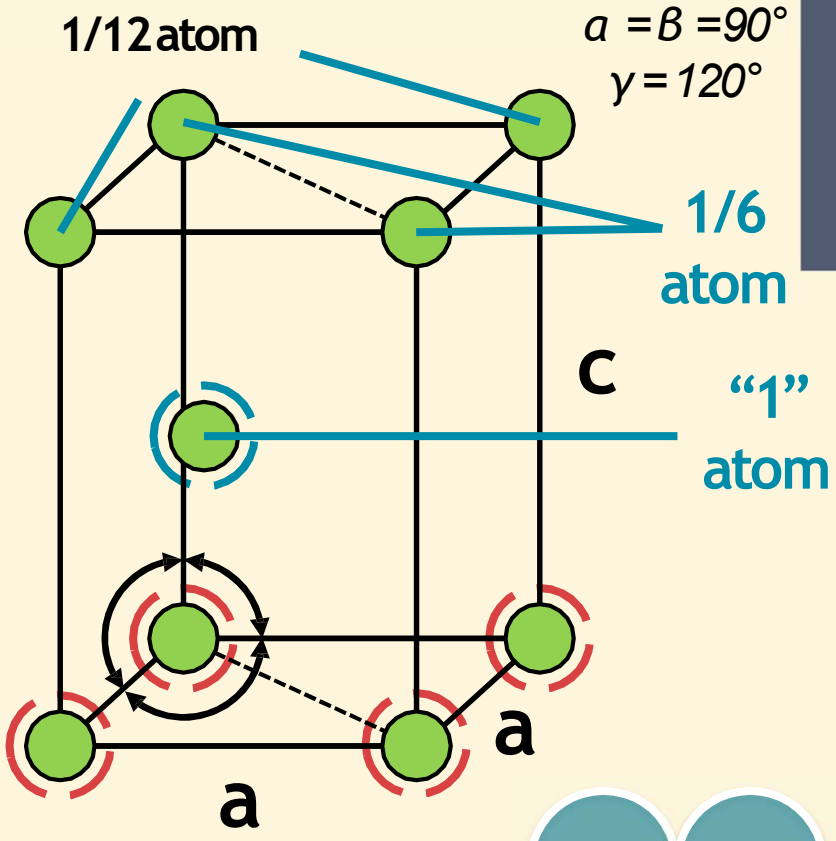
$$V_{atoms} = \frac{4}{3}\pi R^3$$

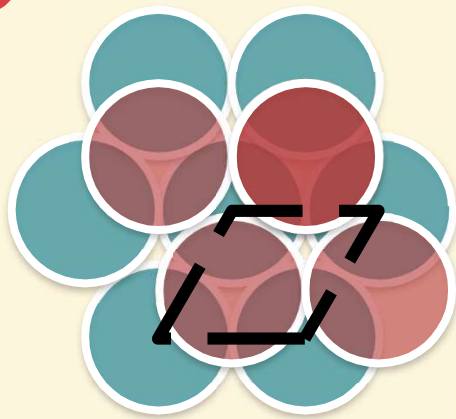
$$V_{unit\ cell} = a^2 c \cos 30°$$

$$a = 2R;\ c = 4\sqrt{\frac{2}{3}}R$$

$$\Rightarrow V_{uc} = (?)(?)(?)$$

$$\therefore V_{uc} = ???R^3$$





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Atomic packing factor – hcp (6)

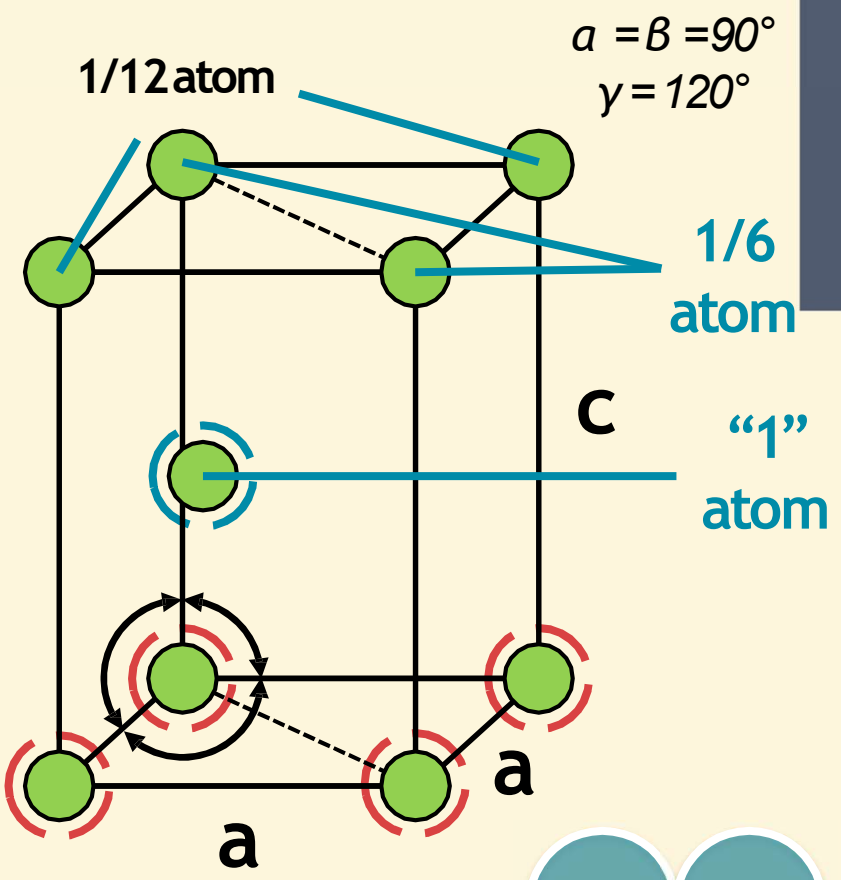
$$N_{atoms} = 2$$

$$V_{atoms} = \frac{4}{3}\pi R^3$$

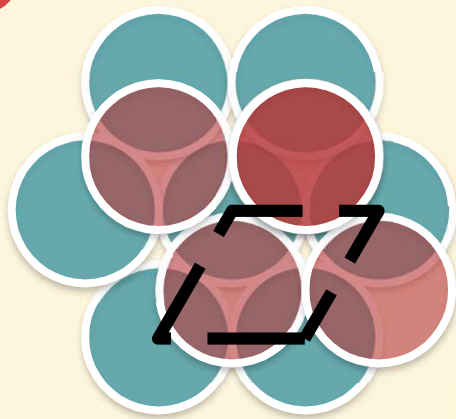
$$V_{uc} = 8\sqrt{2}R^3$$

$$APF = \frac{N_{atoms}V_{atoms}}{V_{unit\ cell}} = \frac{?\left(\frac{4}{3}\pi R^3\right)}{?}$$

$\therefore$ $APF = ???$







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Tips for calculating APF for hcp

- Work from **one-third (2 atoms)** of the total unit cell (6 atoms)
  - The central atom overlaps this reduced cell, but can still be considered as a whole as parts of external atoms intrude in

- To calculate lattice parameter ***c*** from geometry, **remove all unnecessary atoms and bonds** (leaving 4 atom positions only) recognising that half of the remaining base forms an equilateral triangle

- Remember that **base atoms are touching** ($a = 2R$) and that **base atoms touch the central atom**

Summary

- The packing of crystalline structures influences behaviour– the **atomic density** and **symmetry** are key factors

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Concepts and terminology

- Atoms:
  - Substitutional
  - Interstitial
- Unit cell
- Crystal
- Sub-grain
- Grain

- Crystallographic structure:
  - Body-centred cubic (bcc)
  - Face-centred cubic (fcc)
  - Hexagonal close-packed (hcp)

- Atomic packing factor